

RESEARCH ARTICLE

Simulating effects of fitness and dispersal on the use of Trojan sex chromosomes for the management of invasive species

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Abstract

1. The use of Trojan Y chromosomes (TYC) for controlling invasive species involves manipulating the sex chromosomes of captive-raised individuals. Following release, the offspring of these individuals consist of only one sex, thereby skewing the sex ratio of the invasive population and potentially leading to eradication. Simulation models are needed that can inform managers about how to maximize the likelihood of invasive species eradication, since implementation of this novel management approach in the field is still rare.
2. Here, we present the first spatially explicit, mechanistic simulation model of a real-world TYC program for invasive species eradication. Using a brook trout *Salvelinus fontinalis* system model, we investigated the effects of competitive and reproductive fitness of the captive-raised YY males, dispersal behaviour upon their release and landscape heterogeneity on eradication success.
3. Likelihood of eradication was dependent on both the competitive and reproductive fitness of the Trojan individuals. Competitive fitness (i.e. survival) had a higher threshold for eradication, below which the invasive populations were not eradicated.
4. Movement ecology of both the wild and YY male populations was important for eradication. Under a restricted dispersal scenario for YY males following their release, the wild population was not extirpated but maintained a stable, yet reduced, population size. Analysis of landscape configuration indicated that time to eradication of local patches increased with greater connectivity within the stream network.
5. In addition to sex ratio distortion, density-dependent mortality resulting from outplantings made an important contribution to population decline and therefore may also affect native competitors.
6. *Synthesis and applications.* The use of Trojan sex chromosomes to skew population sex ratios is a novel method for the suppression and eradication of aquatic invasive species. Results from our modelling work indicate that while eradication is possible, maximizing its likelihood requires an understanding of the fitness and movement ecology of both the wild and YY male populations of the invasive species. Both our model and the principles derived from this study related to fitness

and behavioural landscape ecology can be broadly applied to other invaded species and systems.

KEYWORDS

behavioural landscape ecology, brook trout, CDMetaPOP, fisheries management, relative fitness, *Salvelinus fontinalis*, simulation modelling, YY males

1 | INTRODUCTION

Aquatic invasive species (AIS) present major challenges for the conservation and management of biodiversity and natural resources world-wide (Gallardo, Clavero, Sánchez, & Vilà, 2016; Havel, Kovalenko, Thomaz, Amalfitano, & Kats, 2015). Invasions in aquatic environments can be particularly difficult to prevent and manage because of how easily AIS can be moved among bodies of water and how susceptible aquatic environments are to natural and anthropogenic change (Havel et al., 2015; Rahel & Olden, 2008). Non-native fish species, for example, have a history of being transported and introduced to new environments for recreational purposes (Gozlan, Britton, Cowx, & Copp, 2010). Other species are incidentally transported between bodies of water in ballast containers or attached to the hulls of boats (Bailey, 2015; Clarke Murray, Pakhomov, & Therriault, 2011). As a result, AIS invasions threaten biodiversity and can be detrimental to ecosystem functions and services (Charles & Dukes, 2008; Molnar, Gamboa, Revenga, & Spalding, 2008).

The use of genetic tools to control invasive species (i.e. genetic pest management, GPM) includes a number of promising techniques for the control of AIS populations without the financial and ecological cost of other widely used methods such as manual or chemical removal (Cotton & Wedekind, 2007a; Harvey-Samuel, Ant, & Alphey, 2017; Simberloff, 2009). These tools are being developed for the control of a wide variety of taxa including insects and all classes of vertebrates (Harvey-Samuel et al., 2017). One approach to GPM, termed sex-ratio distortion, consists of manipulating the genotype of captive individuals such that following release, their offspring consist of only one sex. For example, in an XX/XY chromosomal system, captive males are produced with two Y chromosomes so that all offspring of these 'Trojan' YY supermales (M_{YY}) are also male. This is accomplished by first feminizing XY males using estradiol, and then treating their progeny to produce both YY males and YY females. These fish are then crossed to produce the YY male broodstock for release (Schill, Heindel, Campbell, Meyer, & Mamer, 2016). The goal of Trojan Y chromosome (TYC) programs is to skew the sex ratio of the wild population, eventually culminating in the eradication of the AIS over time. This system has recently been developed for several aquatic species for the eradication of AIS (Jiang et al., 2018; Kennedy, Meyer, Schill, Campbell, & Vu, 2018; Thresher et al., 2014).

The effectiveness of a TYC program may be limited by several factors (Cotton & Wedekind, 2007b; Stelkens & Wedekind, 2010). For example, the relative fitness of the YY males may limit their ability to reproduce and pass on the Trojan chromosome if fitness

is lower than their wild counterparts (Stelkens & Wedekind, 2010). Reductions in fitness may be a result of an accumulation of deleterious alleles or degeneration of the Y chromosome (Charlesworth & Charlesworth, 2000). Indeed, YY genotypes can be lethal for some fish species and reduced viability has been demonstrated for others (Devlin & Nagahama, 2002; George, Thavamani, & Kavumpurath, 1994). For salmonids, evolution of the Y chromosome appears to be relatively recent, with X and Y chromosomes being mostly homogeneous other than the sex-determining gene (Yano et al., 2013). While there is potential for suppressed recombination in some species (e.g. Atlantic salmon; Kijas et al., 2018), fitness implications for most species are unknown (but see Kennedy et al., 2018). In addition to genetic factors, reductions in fitness may also occur as a result of being raised in a captive environment, where conditions vary substantially from the wild (Araki, Berejikian, Ford, & Blouin, 2008). Regardless of the source, a fitness differential may manifest in terms of competitive fitness, (i.e. the ability to compete for resources such as food and habitat; Wiley, Whaley, Satake, & Fowden, 1993) or reproductive fitness as a result of physiological or behavioural mechanisms that limit the reproductive success of the YY males (Aikio, Valosaari, & Kaitala, 2008; Christie, Ford, & Blouin, 2014). Implementing a TYC program without an understanding of the likely effects of relative fitness may limit the program's ability to skew the sex ratio and drive the invasive species to eradication.

Another factor likely to affect the success of a TYC program is the initial dispersal behaviour of the YY males. Logistical constraints could limit the likelihood of introducing YY males evenly throughout an invaded area. Thus, managers may be dependent on the dispersal of the introduced individuals to achieve widespread sex-ratio distortion. Most animal dispersal distributions are leptokurtic and long-distance movements away from release sites tend to be rare (Clobert, Baguette, Benton, & Bullock, 2012). Indeed, the majority of literature examining post-release movements of hatchery-reared salmonids indicates relatively sedentary dispersal behaviour (e.g. Helfrich & Kendall, 1982; Symons, 1969).

Questions about the effects of factors such as relative fitness or dispersal distributions can be explored using simulation models prior to implementation in the field. Gutierrez and Teem (2006) produced the first mathematical model of a TYC program for fish, demonstrating that such a program would theoretically lead to eradication of the invasive species. Subsequent mathematical and simulation models have explored factors such as relative fitness, mating preference and management strategy (Aikio et al., 2008; Gutierrez, Hurdal, Parshad, & Teem, 2012; Schill, Meyer, & Hansen, 2017). However, in a review

of genetic control of invasive species, Harvey-Samuel et al. (2017) indicated a need for methods capable of capturing the complex feedback mechanisms inherent to ecological systems. Such models are especially needed for systems that might be unpredictable under extreme conditions such as low densities (Zipkin, Kraft, Cooch, & Sullivan, 2009). Spatially explicit models, which simulate real-world heterogeneity across landscapes, can elucidate the effect of landscape heterogeneity and connectivity on ecological processes. For example, an invasion that consists of highly connected populations may be more difficult to eradicate than one that consists of several isolated populations (Minor, Tessel, Engelhardt, & Lookingbill, 2009; Stewart-Koster, Olden, & Johnson, 2015).

Here, we present a mechanistic, spatially explicit model to simulate the control of an invasive species through a TYC program using principles of individual-based modelling and behavioural landscape ecology. Specifically, we simulated the release of YY male brook trout into an established population of non-native wild brook trout that compete with native westslope cutthroat trout *Oncorhynchus clarkii lewisi*. Our goal was to understand how time to eradication and/or minimum observed population size might be affected by (a) the relative competitive and reproductive fitness of YY male fish; (b) the dispersal distribution of YY male fish and (c) the connectivity of the river network. Our model is mechanistic in that we simulated the

life history, behaviour and interactions of individual fish across a spatially explicit landscape modelled on a real-world system. Because our model can be adapted to simulate TYC programs targeting the eradication of most undesirable species, the principles derived from this study are broadly applicable to other non-native species and the systems they invade.

2 | MATERIALS AND METHODS

2.1 | Study system

The simulation extent of our study comprised the Sullivan Creek watershed, a major tributary to the lower Pend Oreille River in north-eastern Washington State, USA (Figure 1). The principal trout species inhabiting Sullivan Creek include native westslope cutthroat trout and non-native brook trout. Brook trout were stocked extensively in tributaries to the lower Pend Oreille River from 1933 until 1981 (Kloempken, 1996), and while they are technically managed as a non-native game species rather than an invasive species in this area, recent management plans have targeted their suppression and eradication in areas critical to native trout species. Brook trout are present throughout most of the mainstem of Sullivan Creek and in the lower portions

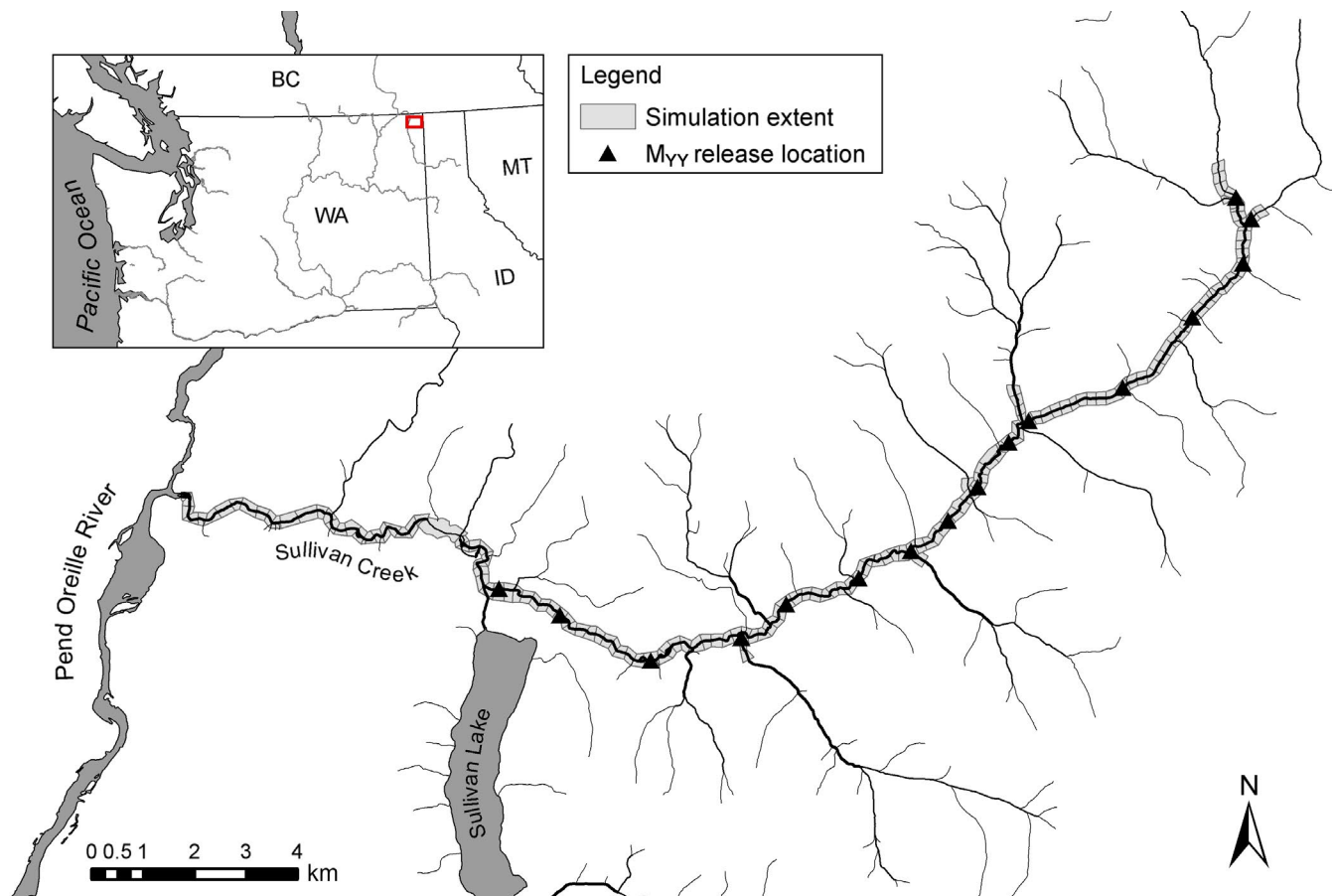


FIGURE 1 Simulation extent and individual patches for the brook trout Trojan Y chromosome model in the Sullivan Creek Watershed in northeastern Washington State, USA. Light gray polygons represent the model patches and the current range of brook trout in Sullivan Creek. Triangles indicate release locations for YY male brook trout for simulations that tested dispersal capacity

of some tributaries near their confluence with the mainstem, occurring in sympatry with westslope cutthroat trout throughout their distribution within the system. Brook trout distribution, however, does not extend as far upstream as that of westslope cutthroat trout in either the mainstem or tributaries due to physical stream barriers that preclude upstream movement. In mainstem habitats where the two species overlap, brook trout represent between 17% and 81% of the combined abundance of the two species, and generally outnumber Westslope cutthroat trout in side channel habitat (R.K. Simmons, unpubl. data).

2.2 | Model description

To simulate population dynamics, movement and strategies for brook trout control, we used the software CDMetaPOP version 1.39 (Landguth, Bearlin, Day, & Dunham, 2017; Mims et al., 2019). We built upon a system model that was previously developed to simulate electrofishing suppression activity on brook trout populations (Day, Landguth, Bearlin, Holden, & Whiteley, 2018). Following, we provide only broad descriptions of CDMetaPOP processes except where modifications have been made to the original model. For detailed descriptions of the original brook trout system model and of CDMetaPOP, see Day et al. (2018) and the CDMetaPOP user manual (<https://github.com/ComputationalEcologyLab/CDMetaPOP>), respectively. A full table of parameter values is available in Supporting Information.

CDMetaPOP is an individual-based, demogenetic modelling framework that simulates movement of individuals and exchange of genetic information across heterogeneous landscapes that consist of discrete patches (i.e. subpopulations). Movement of individuals (e.g. dispersal, mating) between patches occurs based on a draw from a probability distribution developed by calculating pairwise resistance-to-movement values between patches. Each year, individuals undergo temperature-based growth, maturation, mating, reproduction, dispersal, and both density-dependent and density-independent mortality. In the present application of CDMetaPOP, most of these processes were parameterized based on empirical data collected to inform a long-term project to suppress brook trout and restore native salmonid populations.

2.3 | Implementation of population introductions and Trojan Y chromosomes

To simulate the use of Trojan YY males for control of the brook trout population, we developed a new module within CDMetaPOP that allows for the introduction of new populations at the beginning of any time step (i.e. year) during the simulation. Introduced populations can be distributed in any spatial configuration desired by the user to mimic management strategies. Newly introduced populations are assigned an independent set of parameters governing processes such as mortality, maturation, fecundity and movement. We also implemented the ability to assign sex ratios to both the initial and

introduced populations with three options for sex chromosomes: XX, XY and YY. Following mating, sex chromosome inheritance followed Mendelian segregation. This module enabled us to simulate the release of individual YY male brook trout into an existing wild brook trout population.

To simulate fitness of YY male fish relative to wild-type fish (i.e. competitive fitness and reproductive fitness), we implemented a measure of hybridization termed the hybrid index (HI; Allendorf, Leary, Spruell, & Wenburg, 2001; Nathan et al., 2019). While originally developed to explore fitness consequences of interspecific hybridization, this index can be applied to quantify fitness of a unique genotype within a species. All wild-type individuals were assigned an HI value of 1, and all YY males an HI value of 0. HI values of offspring were calculated by averaging those of the parents. In the simulations, we followed the Dominant-Preference hybridization model of M'Gonigle and FitzJohn (2010) so that hybrids were indistinguishable from wild-type fish in terms of fitness (i.e. only hatchery-reared fish had reduced fitness). To implement effects of competitive fitness between wild-type fish and YY male fish, we supplied a value of relative fitness between 0 and 1, which represented the relative probability of surviving density-dependent mortality for all fish where $HI = 0$ (i.e. M_{YY}). This mechanism represented the relative ability of hatchery-reared YY male brook trout to compete for resources with wild brook trout. To simulate differential reproductive fitness, we utilized CDMetaPOP's assortative mating module and again implemented the HI index and Dominant-Preference hybridization model. As before, we supplied a value between 0 and 1 that represented the probability of a YY male brook trout mating relative to a wild male brook trout (Kennedy et al., 2018). This implementation of fitness assumes independence between competitive and reproductive success, although in reality the two may not be independent (e.g. size may influence both competitive and reproductive success). However, given that size-specific competition occurs by default within the density-dependent mortality module and that we are simulating competition between hatchery-reared fish and wild fish, this model of fitness highlights the behavioural mechanisms that may result in naïve hatchery fish being outcompeted by wild fish (e.g. inability to forage or successfully spawn). Also of note is that the two different fitness values interact in a multiplicative manner, so that two fitness values of 0.5 likely interact to yield an overall fitness of 0.25.

2.4 | Baseline simulation scenario

We divided the occupied extent of our study area into 152 stream reaches (i.e. patches), each approximately 200 m in length and corresponding to the sampling units utilized in ongoing brook trout suppression (Figure 1). We created a baseline scenario against which to compare alternative treatment and fitness scenarios. This baseline scenario included manual brook trout suppression (see Day et al., 2018) in conjunction with annual releases of approximately 9,000 hatchery-reared YY male brook trout of fingerling size. These treatments followed the current schedule for management efforts

planned for the Sullivan Creek watershed, which also included barrier removals and habitat restoration. Treatments continued for 100 years, or until the female portion of the population was eradicated. For this baseline scenario, YY male fitness (both competitive and reproductive) was assumed to be equal with that of wild brook trout, and YY male fish were outplanted in proportion to the density of wild brook trout occupying a given patch. We ran 10 replicates of this baseline scenario and calculated 95% confidence intervals (mean $\pm$ SD * 1.96) for all output metrics including population size over time, sex ratio over time, patch occupancy over time and time to eradication. Confidence intervals reflect variability due to the model's stochastic elements.

2.5 | Alternative treatment scenarios

To demonstrate the effects of each individual treatment on the wild brook trout population, we executed 10 replicates of several alternative treatment scenarios: (a) no treatment, (b) manual suppression only, (c) M_{YY} releases with no suppression, (d) M_{YY} releases with suppression, (e) M_{XY} releases with no suppression and (f) M_{XY} releases with suppression. All other parameters were held the same as in the baseline scenario. Simulating the release of M_{XY} hatchery fish acts as a control to demonstrate the effect of overpopulation on density-dependent mortality in the absence of sex ratio manipulation via Trojan chromosomes.

2.6 | YY male relative fitness scenarios

To test for effects of YY male fitness on eradication of wild brook trout, we ran 100 versions of the baseline scenario while varying the competitive fitness and reproductive fitness of YY male fish in a bivariate sensitivity analysis (Table 1). To sample values between 0 and 1 for each fitness parameter, we used a Latin hypercube sampling (LHS) design (Seaholm, Ackerman, & Wu, 1988). In LHS, the range of values for each parameter is divided into N subsections which are each then sampled orthogonally ($N = 100$). This design minimizes correlation among input parameters, maximizing the diversity of parameter configurations and achieving inference more efficiently

than through random sampling. To compare the relative success of each simulation scenario in eradicating wild brook trout, we created an Eradication Index. This index represented a combination of the relative success of a simulation in terms of time to eradication or—in the case where the population persisted—minimum number of females observed and is calculated between 0 and 1 as

$$\text{Eradication Index} = 1 - \left[\left(\frac{T_{\text{obs}} - T_{\text{min}}}{T_{\text{max}} - T_{\text{min}}} + \frac{F_{\text{obs}} - F_{\text{min}}}{F_{\text{max}}} \right) / 2 \right]. \quad (1)$$

Here, T_{obs} , T_{min} and T_{max} are the observed, minimum and maximum time to eradication in years across the 100 parameter scenarios respectively. F_{obs} , F_{min} and F_{max} refer to the observed, minimum and maximum number of females, respectively, from the best and worst performing scenario. In the case where a simulation failed to achieve extirpation, $T_{\text{obs}} = 100$. In the case where a population was extirpated, $F_{\text{obs}} = 0$. To analyse effects of fitness on eradication, we used multiple linear regression and produced contour plots for visualization. All analyses were performed in R v 3.5 (R Core Team, 2018).

2.7 | YY male initial release dispersal scenarios

Because initial movement of hatchery fish in this system is unknown, we tested for effects of the initial dispersal distribution of YY males away from release locations on eradication success. For these simulations, we modelled a realistic management scenario in which YY males were outplanted to 16 locations in the Sullivan Creek watershed. Release locations were strategically chosen based on wild brook trout densities and proximity to each other. No release locations were assigned to the lower portion of Sullivan Creek (Figure 1) because brook trout occurred at very low densities in these reaches. We conducted simulations using four alternative movement distributions (Random, Gaussian 1, Gaussian 2 and Pareto; Table 1), and analysed the final outcomes using ANOVA. The Pareto distribution represented the most restricted dispersal scenario (Rodriguez, 2002), followed by the intermediate Gaussian scenarios, with the random distribution representing the greatest dispersal. For the random distribution, dispersing hatchery fish moved to a randomly selected patch regardless of

TABLE 1 Factors varied across simulations of brook trout control treatments in the Pend Oreille River watershed in northeastern Washington State

Factor	Levels	No. of simulations	Notes
Competitive fitness	Continuous, 0–1	100	Implemented during density-dependent mortality
Reproductive fitness	Continuous, 0–1	100	Implemented via assortative mating among wild-type fish
YY dispersal distribution	Random, Gaussian 1, Gaussian 2, Pareto	10 reps per scenario	Two Gaussian distributions were tested with a mean of 0 km ($SD = 0.5$ km) and a mean of 0.5 km ($SD = 2.6$ km; Baird, Krueger, & Josephson, 2006). The Pareto distribution shape and scale were 1.346 and 202.5 respectively (Letcher et al., 2015)

cost distance. The other three scenarios assigned movement based on a random draw from the respective dispersal distance distribution. Hatchery fish dispersed according to the same distribution as wild fish following their first spawning season in Sullivan Creek.

2.8 | Landscape factors

We tested whether landscape connectivity and spatial heterogeneity in brook trout densities affected the amount of time a given patch was occupied. For connectivity, we used the *igraph* package in R to conduct network analysis on the landscape and compute eigenvector centrality scores for each patch (Bonacich, 2007; Csardi & Nepusz, 2006). We then used regression analysis on results from 10 replicates of the baseline scenario to test how well the natural log of the centrality score and carrying capacity of each patch predicted the proportion of time steps a given patch was occupied during a simulation. In other words, we tested whether the relative location and carrying capacity of a patch affected that patch's time to eradication.

2.9 | Global sensitivity analysis

To understand the effect of all other relevant model parameters, we tested the global sensitivity of population size and the Eradication Index to 21 parameters using the same LHS design mentioned above (Seaholm et al., 1988). Parameter ranges were based on estimates from the literature and from field sampling $\pm 10\%$ (à la Schill et al., 2017).

To test and visualize the sensitivity of model outputs to each sampled parameter, we used the 'src' function in the *SENSITIVITY* package in R v 3.5 (Pujol, looss, & Janon, 2017; R Core Team, 2018). This function uses bootstrap replication of output data to produce standardized regression coefficients (SRC) for each tested model parameter (e.g. Nathan, Kanno, & Vokoun, 2017). For our analysis, we used 10,000 bootstrap replicates and a 95% confidence interval. We considered an SRC significant when its confidence interval did not overlap zero.

3 | RESULTS

3.1 | Baseline scenario and alternative treatments

For the baseline model scenario, where YY males had equal fitness with wild males, and YY males were distributed evenly throughout the watershed, mean time to eradication was 28.9 (95% CI ± 6.0) years from the initial suppression treatment (Figure 2 top panel). Mean proportion of patches occupied declined monotonically over time in an inverse sigmoidal fashion (Figure 2 bottom panel). Results from alternative treatment scenarios indicated that suppression alone was the least effective treatment option, failing to result in eradication (Figure 3; Day et al., 2018). YY male releases in the absence of suppression required approximately 80 more years to achieve eradication than the combination of both treatments. Simulations of

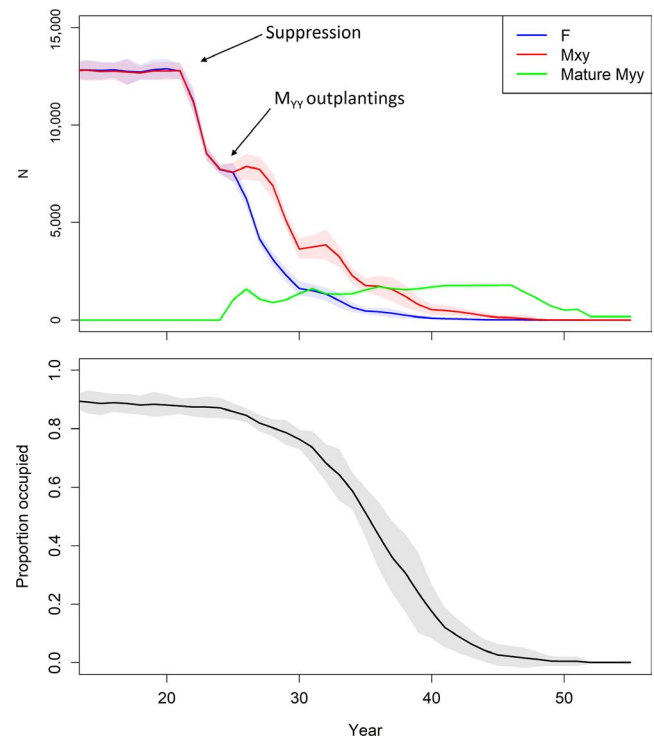


FIGURE 2 Population dynamics over time of brook trout in a simulation model of suppression and eradication via electrofishing and sex chromosome manipulation. Suppression begins in year 23, outplantings of YY male brook trout (M_{YY}) begin in year 26 and both treatments continue until eradication occurs. The top panel displays number of individuals with each chromosomal arrangement, and the bottom panel displays proportion of patches occupied over time. Polygons represent 95% confidence bands

the outplanting of M_{XY} rather than M_{YY} demonstrated that density-dependent mortality and replacement of wild fish by hatchery fish account for a significant proportion of the decline of the wild population (Figure 3).

3.2 | YY male relative fitness scenarios

Both competitive fitness and reproductive fitness were significant predictors of the Eradication Index for a given simulation scenario ($\beta = 0.73$ and 0.56 respectively; $p < 0.001$; $R^2 = 0.88$; Figure 4). Contour plots revealed additional details about how the Eradication Index, time to eradication and minimum number of females interacted with fitness. Slopes of contour lines indicated that the Eradication Index varied most in response to changes to higher values of competitive fitness and lower values of reproductive fitness (Figure 4a). More specifically, a threshold was observed for competitive fitness at a value of approximately 0.6, below which no scenario experienced eradication, regardless of the reproductive fitness value (Figure 4b). No such threshold was clearly observable for reproductive fitness, as some scenarios with very low values of reproductive fitness (< 0.25) and high competitive fitness still experienced eradication

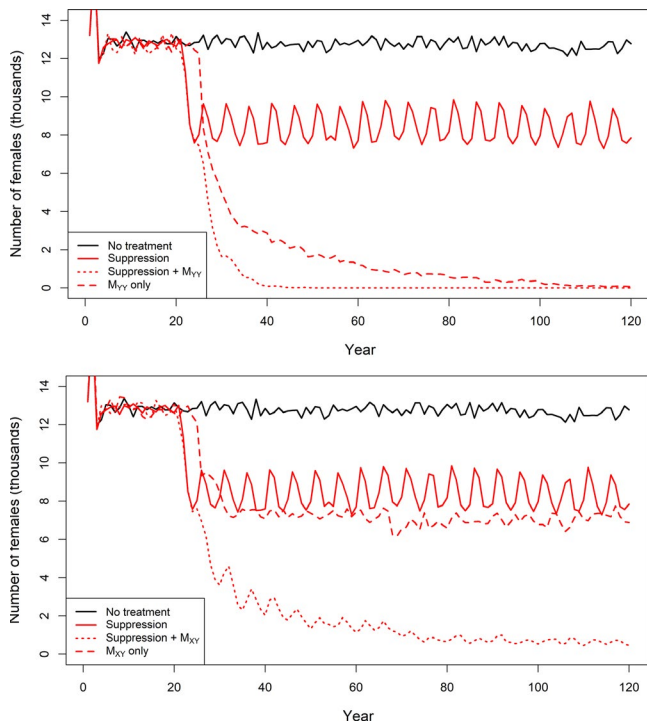


FIGURE 3 Population dynamics over time of the wild brook trout population in a simulation model of alternative control treatments using electrofishing suppression and sex chromosome manipulation. The top panel displays treatments using M_{YY} supermales to skew the sex ratio, while the bottom panel displays the same treatments but with M_{XX} males. M_{XX} releases were simulated to demonstrate effects of density-dependent mortality and replacement of wild fish by M_{YY} fish

(Figure 4b,c). For those simulations where eradication did not occur, contour lines indicated that the minimum number of females observed appeared more responsive to changes in values of reproductive fitness than competitive fitness (Figure 4c).

3.3 | YY male dispersal scenarios

None of the four dispersal scenarios tested (Random, Gaussian 1, Gaussian 2 and Pareto) consistently resulted in the complete eradication of the wild brook trout population (Figure 5). Four of 10 replicates of the Gaussian 1 scenario and one replicate of the Random scenario resulted in extirpation. The Pareto distribution (the most restricted dispersal scenario) performed significantly worse than the others in reducing the brook trout population ($F_{3,36} = 3,234$, $p < 0.001$). Indeed, simulation results indicate that under this restricted dispersal scenario, the wild brook trout population could persist indefinitely at a reduced, stable state.

3.4 | Landscape factors

The proportion of time that a patch was occupied by wild brook trout females was affected by a patch's centrality and carrying

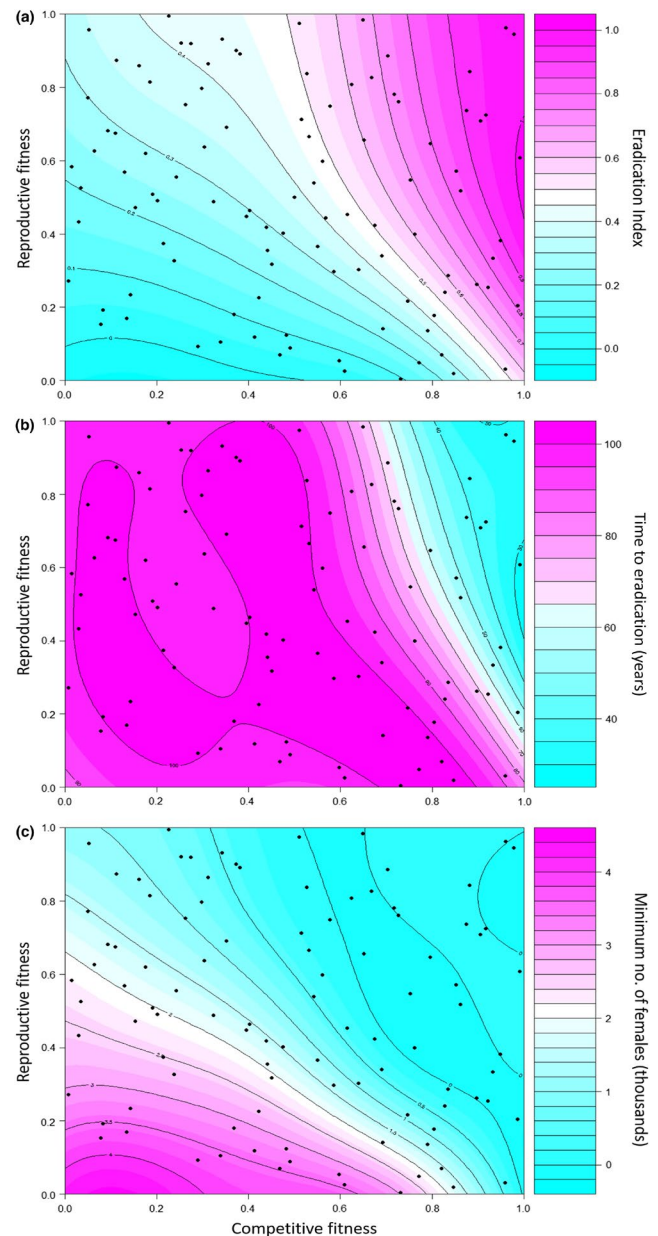


FIGURE 4 Contour plots displaying relationships between success of a wild brook trout eradication program and fitness of hatchery YY male brook trout relative to wild brook trout. Each dot represents a single simulation scenario and associated values of competitive fitness and reproductive fitness. Response variables include (a) the Eradication Index, (b) time to eradication and (c) the minimum number of females observed. In the case that eradication did not occur, that simulation was assigned a value of 100 years. Competitive fitness indicates the ability of YY male brook trout to compete for resources and survive density-dependent mortality. Reproductive fitness indicates the YY male brook trout's relative ability to compete with wild male brook trout for mates

capacity ($\beta = 0.0089 \pm SE 0.0010$ and 0.00043 ± 0.000077 respectively; $p < 0.001$; $R^2 = 0.60$). Patches more central to the network and with higher carrying capacity were occupied more often than patches less central or with lower carrying capacity, despite the

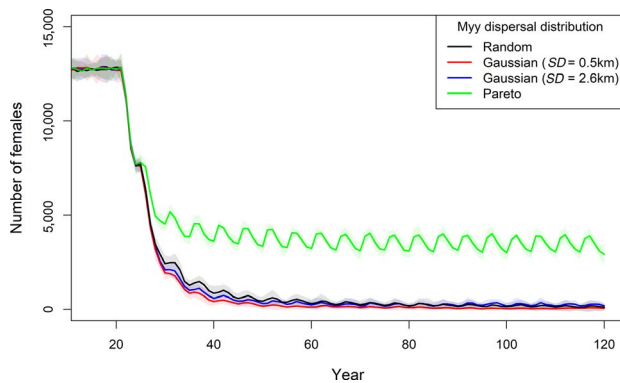


FIGURE 5 Relationship between initial dispersal distribution of YY male brook trout and success of a wild brook trout eradication program with 95% confidence bands. Simulations were set in the Sullivan Creek Watershed in northeastern Washington State, USA

number of YY fish released being proportional to a patch's carrying capacity.

3.5 | Global sensitivity analysis

Stable population size was most sensitive to parameters associated with annual reproduction and recruitment, as the slope and intercept of the fecundity equation as well as annual egg mortality all had significant regression coefficients ($\text{SRC} = 0.42 \pm 95\% \text{ CI } 0.045, 0.28 \pm 0.037, -0.49 \pm 0.044$ respectively; Figure 6a). Stable population size was also sensitive to carrying capacity ($\text{SRC} = 0.33 \pm 0.044$) and to several factors associated with temperature and growth. Population size was not sensitive to changes in movement rates and distributions or other growth parameters.

Success of the TYC program as measured by the Eradication Index was similarly most sensitive to changes to the slope and intercept values in the equation regulating fecundity ($\text{SRC} = -0.61 \pm 0.049$ and -0.49 ± 0.044 respectively; Figure 6b). The Eradication Index was less sensitive than population size to changes to temperature, growth and egg mortality. Unlike stable population size, the Eradication Index was sensitive to the slope and intercept regulating maturation timing ($\text{SRC} = -0.21 \pm 0.037$ and -0.24 ± 0.037 respectively) and insensitive to changes in carrying capacity ($\text{SRC} = -0.022 \pm 0.035$). The Eradication Index was also sensitive to the amount of suppression effort (i.e. electrofishing mortality rate) implemented in the management plan ($\text{SRC} = 0.15 \pm 0.036$).

4 | DISCUSSION

The primary goal of this paper was to produce a mechanistic simulation model of a TYC program to inform management of invasive species. We used this model to explore how relative fitness, dispersal behaviour and landscape connectivity affected the success and time required for successful eradication. This is the first model of the TYC management strategy that is individual-based

and spatially explicit, simulating a real-world riverscape and its invasive population. Furthermore, this is the first TYC simulation model to be grounded in empirical data derived from the managed invasive population and its environment, including population demographics, genetics, heterogeneity in population density, landscape connectivity and temperature. Thus, our study provides novel contributions to the existing modelling work on TYC programs, specifically that it is (a) mechanistic (i.e. bottom-up) and (b) spatially explicit.

Earlier mathematical simulation models have laid the theoretical foundation for the use of Trojan sex chromosomes to eradicate invasive species. Despite excluding suppression mortality, these studies have indicated that complete eradication of the invasive species is possible (e.g. Gutierrez & Teem, 2006; Parshad, Kouachi, & Gutierrez, 2013). These models have generally employed a top-down approach to simulation through the use of differential equations governing factors such as birth rates, death rates, population growth rates and mating. These models have explored the effects on time to eradication of factors such as number of YY fish released (Gutierrez & Teem, 2006), the use of M_{YY} versus F_{YY} fish (Kelly & Wang, 2017), dendritic stream configuration (Gutierrez et al., 2012) and Allee effects (Beauregard et al., 2019).

We employed a bottom-up modelling approach in that we simulated behaviours (e.g. dispersal, mating) and processes (e.g. mortality, growth) associated with individual fish from which population-level patterns emerged. One benefit of this type of mechanistic model is that it can be more likely to capture unpredictable outcomes from a complex system, such as compensatory responses of a fish population at extremely low densities (Day et al., 2018; Meyer, Lamansky, & Schill, 2006; Zipkin et al., 2009). Such models are also more robust to modelling alternative landscapes or management scenarios and more generalizable to other systems and scales (Radchuk, Kramer-Schadt, & Grimm, 2019; Wiens, Chr, Horne, & Ims, 1993). Adding a spatially explicit landscape further advances TYC modelling because it allows for demographic processes, such as population decline, to vary across the landscape rather than occur uniformly. This heterogeneity can be driven by environmental factors associated with an individual's local environment such as temperature, which regulates growth and therefore fecundity. Modelling explicit space also allows for landscape configuration and heterogeneity to affect movement associated with mating, dispersal, and recolonization, potentially affecting the rate of decline.

Our model generally agrees with previous work in demonstrating that eradication of an invasive population is possible using a TYC strategy (Parshad et al., 2013; Schill et al., 2017), and unlikely using manual suppression alone (Day et al., 2018; Meyer et al., 2006). However, our predicted time to eradication under ideal conditions is longer than some previous models have predicted. Furthermore, under less than ideal conditions, complete eradication under a TYC program in addition to manual suppression may require intense effort over a long period of time. Our model also

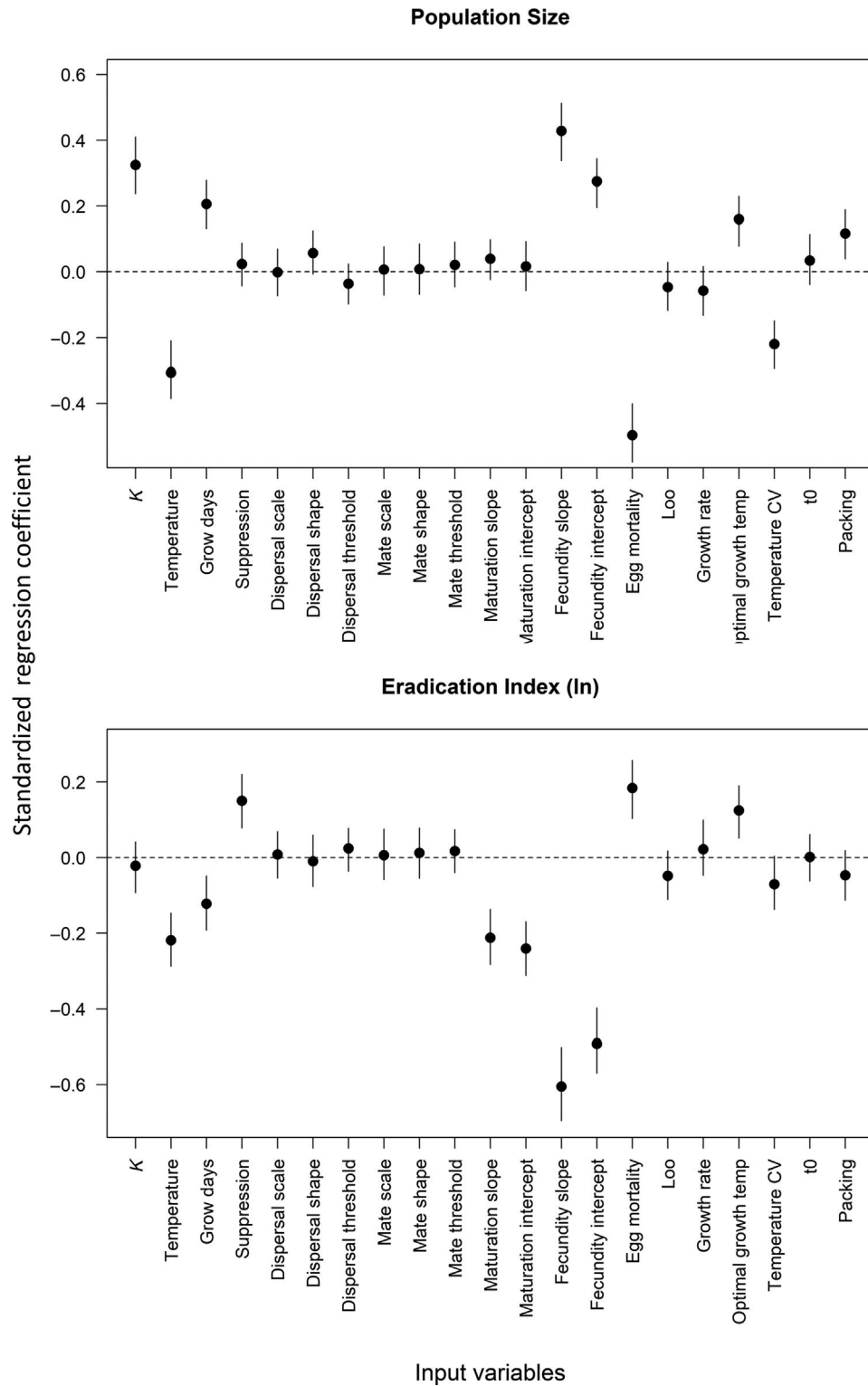


FIGURE 6 Global sensitivity analysis indicating standardized regression coefficients ($\pm 95\%$ Confidence Intervals) for 21 model parameters. One hundred simulations were conducted using a Latin hypercube sampling design. The top panel displays sensitivity of the stable population size of the brook trout populations and the bottom panel displays sensitivity of the natural log of the Eradication Index, which is a measure of how successful the simulation was in reducing the wild brook trout population. K = carrying capacity. Scale, shape and threshold refer to movement distributions for dispersal and mating movement. Loo, Growth rate, Max temp growth, Temperature CV and t0 are parameters used in the Von Bertalanffy equation modified for temperature-dependent growth. Packing regulates density-dependent mortality by size class

demonstrates the importance of density-dependent mortality and replacement of wild fish with YY fish (and not just sex ratio distortion) for precipitating the decline of the wild population. Such effects would be experienced by not only the target species, but competitors of the target species as well. Given the timeframe required for eradication, careful attention should be paid to the effects of a TYC program on native species that compete with the invasive species.

Both competitive fitness and reproductive fitness had a significant effect on the success of, and timeframe required for, eradication. This result underlines the need for managers to understand the relative fitness of hatchery-reared YY males in order to maximize the likelihood of eradication. Senior, Krkosek, and Nakagawa (2013) developed an individual-based model that simulated the use of F_{YY} fish, and found that fecundity of the feminized supermales, not survival or mating probability, was the limiting factor for eradication. Our study, which simulated M_{YY} releases, is more comparable to the age-structured model of Schill et al. (2017) which found that probability of eradication was lower when M_{YY} fitness (combined survival and reproductive success) was reduced. While our results generally agree with those of Schill et al. (2017), our model evaluated the effects of competitive fitness (survival) and reproductive fitness independently. Our results demonstrate the existence of a threshold for relative competitive fitness, below which eradication is unlikely regardless of reproductive fitness. While no clear threshold was observed for reproductive fitness, it did have a significant effect on the rate of population decline and time required for eradication. Overall, a 10% decrease in competitive fitness led to a 7.3% decrease in effectiveness of the TYC strategy (i.e. Eradication Index), while the same decrease in reproductive fitness caused a 5.6% decrease in effectiveness. Complete eradication was much less likely for simulations with low values of competitive fitness than for low values of reproductive fitness. We acknowledge, however, that the range of fitness values we tested may be much wider than that may occur in reality, and therefore long-term monitoring of treated populations is necessary to refine estimates of relative fitness and improve model reliability.

Given that fitness of hatchery-reared fish tends to be lower than that of wild fish (Araki et al., 2008), quantifying competitive and reproductive fitness may be critical to evaluating the likelihood of success of a TYC program. Additionally, steps taken in the hatchery to optimize the fitness of the YY males (e.g. timing of release, size of fish at release, heterozygosity) would increase likelihood of eradication (Araki et al., 2008; Danzmann, Ferguson, & Allendorf, 1989). While improving the competitive ability of the hatchery fish would lead to a more rapid decline of the target population, it may also exacerbate any detrimental impacts on native fish populations. In our study system, westslope cutthroat trout compete with brook trout for resources. Thus, managers outplanting YY males should consider balancing time to eradication with any negative demographic and genetic consequences that may result from competition between the YY male population and the native population (Dunham, Adams, Schroeter, & Novinger, 2002).

Landscape configuration and the ability of animals to move through that landscape are additional factors that can limit the effectiveness of a TYC program (Bélisle, 2005). The only treatment so far of this problem is a mathematical model that explored the theoretical implications of a dendritic riverine system on eradication using a TYC strategy (Gutierrez et al., 2012). Combining principles from both landscape ecology and movement ecology (Lima & Zollner, 1996), our results show that the spatial location of a local population within a landscape can impact the time required for that local population to be eradicated. More connected populations are more likely to receive immigrants and thus are more likely to be recolonized following local extirpation than a more isolated population (Taylor, Fahrig, Henein, & Merriam, 1993). To deal with this phenomenon, managers may consider supplementing centralized locations or locations with high densities of the invasive species with additional treatment.

In addition to movement of the wild individuals, movement of the hatchery-reared YY males is another factor that can affect the success of a TYC program. Because logistical constraints can make it difficult to distribute YY males evenly throughout an invaded area, managers may be dependent on the dispersal ability of the YY males themselves to treat the entire invasive population. We show that under a realistic management strategy, eradication may not occur if post-release movements follow a restricted dispersal distance distribution (e.g. Pareto). In fish populations, less is known about the initial dispersal movements of hatchery-reared fish compared to natural dispersal of wild fish, though most studies indicate a sedentary pattern of behaviour, with rare long-distance movements (Bettinger & Bettoli, 2002; Cresswell, 1981; Helfrich & Kendall, 1982; Jorgensen & Berg, 1991). Such an initial dispersal distribution may result in uneven coverage throughout a stream. In such cases, the lack of interaction with just a few wild individuals in isolated or inaccessible areas could result in repopulation of an entire watershed (Franssen, Davis, Ryden, & Gido, 2014; Meyer et al., 2006). Thus, post-release monitoring of dispersal movements would be critical to informing future management strategy.

Sensitivity analysis revealed several additional parameters associated with the wild brook trout population that may warrant monitoring during implementation of a TYC program. Specifically, parameters associated with an increase in the reproductive capacity of wild individuals (e.g. maturation, fecundity) increased time to eradication. Any compensatory response to suppression mortality associated with these variables could increase the time and effort required for eradication. We note, however, that the parameter space we explored for these variables was $\pm 10\%$, and not necessarily indicative of real-world variation. Varying other more intuitive parameters such as carrying capacity or suppression effort had little to no impact on eradication success. While most studies have focused on the reproductive ability of the YY males (Senior et al., 2013), our results highlight the need to understand the reproductive ability and response of the wild population to suppression as well.

Our mechanistic simulation model of TYC strategies incorporated more realism than previous models, and made advances by simulating

the movement and life history of individual fish across a spatially explicit landscape. This approach incorporated principles from invasive species biology, landscape ecology, behavioural ecology, fisheries biology and computational ecology to produce a model useful to managers of not only the present study system but of other systems as well. Such interdisciplinary efforts are required to address a problem as pervasive and universal as invasive species. For example, landscape configuration and resistance to movement can play a central role in how invasive species spread (Lustig, Stouffer, Doscher, & Worner, 2017; With, 2002). Animal behaviour (e.g. mate selection, dispersal) can drive rates of hybridization between native and invasive species (Kovach et al., 2015). Because our model incorporates spatially heterogeneous landscapes and animal behaviour, it presents an ideal framework for interdisciplinary study of invasion dynamics.

Our study has a number of limitations that present avenues for future research. Similar to previous modelling efforts, little empirical data exist related to outcomes of TYC management programs (but see Kennedy et al., 2018). However, releases of YY male fish into our study site are ongoing, and monitoring data from that work will continue to be incorporated to recalibrate the model to empirical results. This feedback mechanism presents an ideal opportunity for adaptive management, where management practices and models iteratively inform one another to improve overall performance (Walters, 1986). Thus, future work will focus on how empirical data can inform the modelling effort and vice versa. Additional factors affecting TYC success require future study. For example, how Allee effects might impact time to eradication (Beauregard et al., 2019), or how climate and land use change might affect the viability of the invasive population and likelihood of reinvasion. While our study was limited to a single species, further research on the effects of a TYC program on native competitors such as cutthroat trout is also warranted. Because our modelling framework can be adapted to most invasive systems, such questions can be explored beyond the presently described study system, and principles derived from this study related to fitness and movement behaviour are broadly applicable to other species and systems.

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AUTHORS' CONTRIBUTIONS

C.C.D., E.L.L., R.K.S., W.P.B. and A.B. conceived of the study design and model implementation; C.C.D. and E.L.L. led simulation modelling and writing; A.R.W. and P.M.L. contributed to data collection, analysis and interpretation, and provided input on parameterization and implementation of the model. All authors contributed to the original manuscript, revisions and final approval for publication.

DATA AVAILABILITY STATEMENT

Data available via the Dryad Digital Repository <https://doi.org/10.5061/dryad.qv9s4mwb9> (Day, Landguth, Bearlin, Holden, & Whiteley, 2020).

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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